Bauhaus-Universität Weimar

JOHNS HOPKINS UNIVERSITY

From FEM to Neural Operators: Scientific Machine Learning for Computational Mechanics

Post-processing and Error Norms

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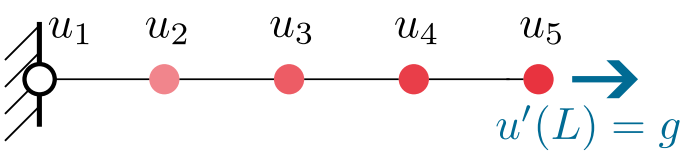
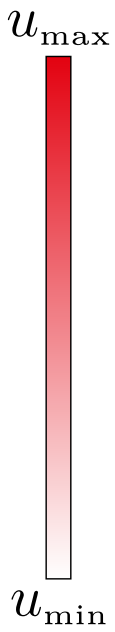
Bauhaus Spring School 2026

What we already have

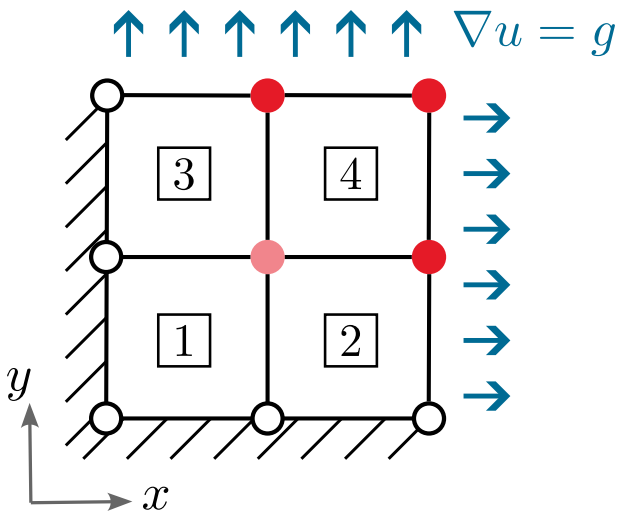
Solution fields:

1D:



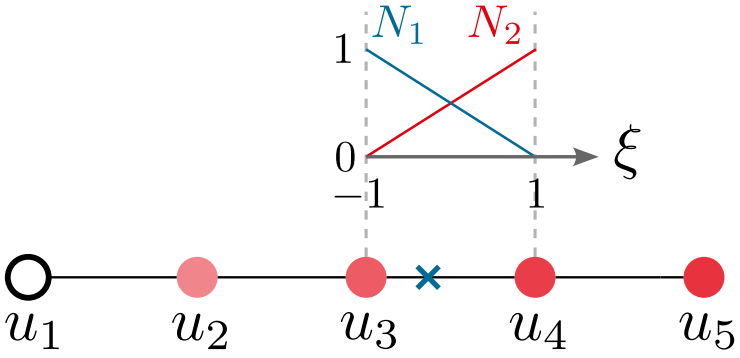


2D:

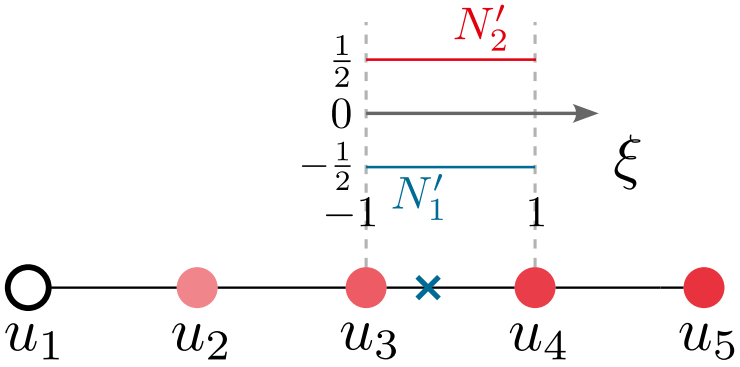


- Nodal solution values **U** obtained from the FEM discretization.
- Next step: Computing u and u' at any point, and error norms.

Interpolation of u and u'



$$u(x(\xi)) = \sum_i N_i(\xi)\, u_i$$

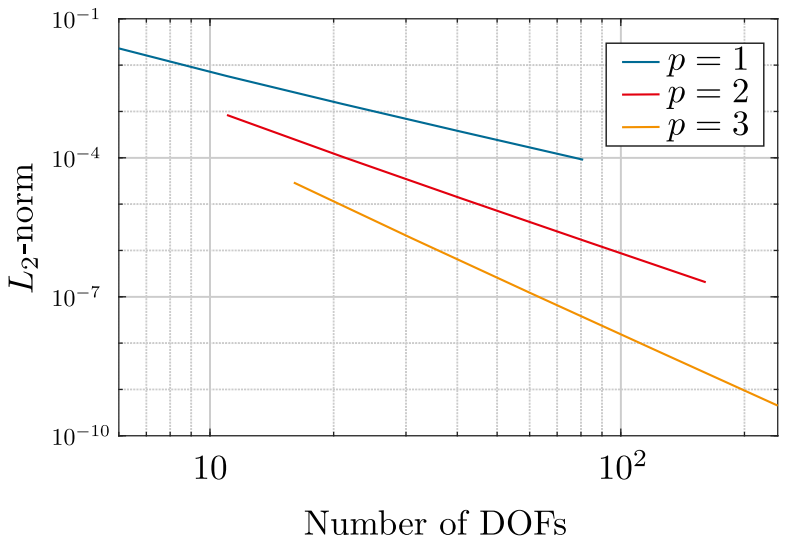


$$u'(x(\xi)) = \sum_i N_i'(\xi)\, u_i$$

Error field and norms

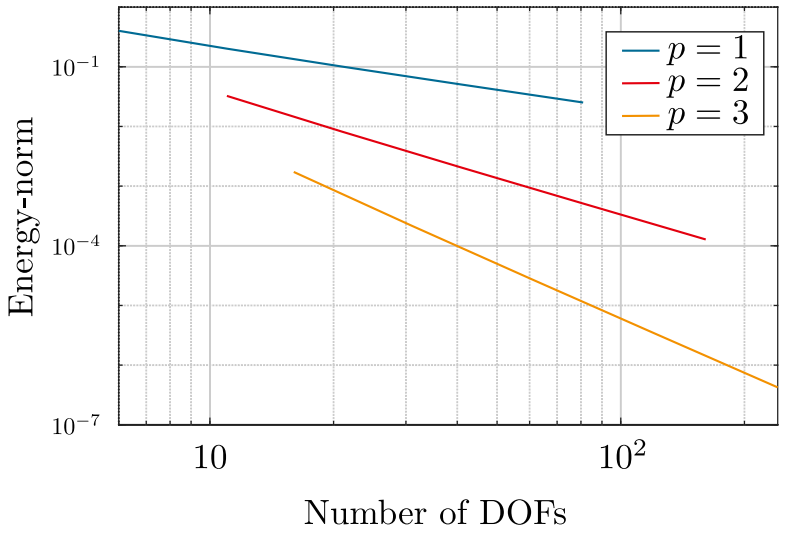
L^2-norm:

$$\|e\|_{L^2(\Omega)} = \left(\int_0^L \left(u(x) - u_h(x)\right)^2 dx \right)^{1/2}$$



Energy-norm:

$$\|e\|_E = \left(\int_0^L \left(u'(x) - u_h'(x)\right)^2 dx \right)^{1/2}$$



Conclusions and outlook

Conclusions

- Post-processing uses the same tools as assembly:
  - Shape functions, gradients, Jacobians, quadrature.
- We can reconstruct $u(x)$ and $u'(x)$ at any point.
- L^2- and energy-norms quantify the error in u and ∇u.

Outlook (PINNs and Neural Operators)

- PINNs: PDE residuals and boundary conditions appear in the loss; error norms can be used for validation.
- Neural Operators: often trained on FEM solution fields (and compared using similar error norms).